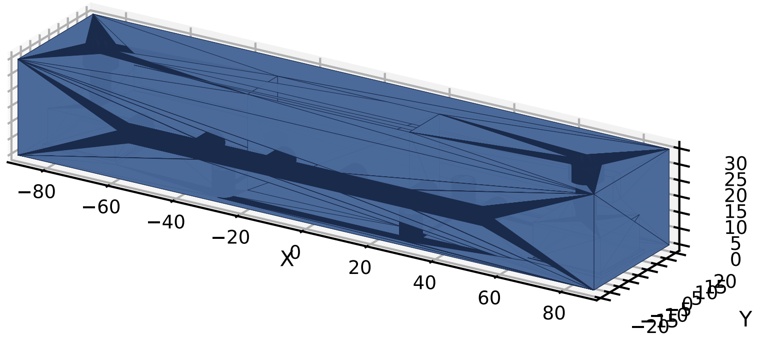
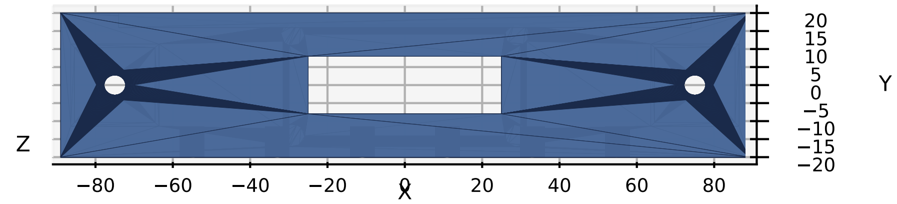
CSM V3 -- HMI Module 10 -- Cross Beam V1.1

*Horizontal cross-member joining mast tops * Carries Pi 4 + future accessories * PETG, 100% infill, 4 walls*

Isometric View



Top View (X-Y) -- mast pockets + Pi inserts

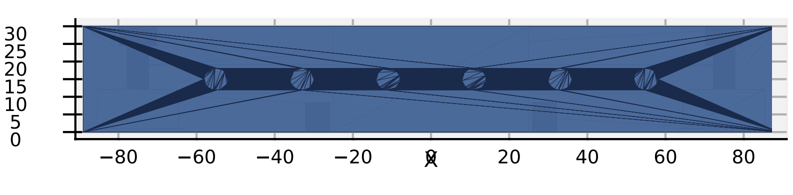


PART	Cross Beam
VERSION	V1.2
MODULE	10 -- HMI
QTY	1
MATERIAL	PETG (PA12 prod)
INFILL	100%, 4 walls
LAYER	0.2 mm
MASS	~216 g
TIME	~8-10 h
INSTALL	top of HMI masts (X=0)

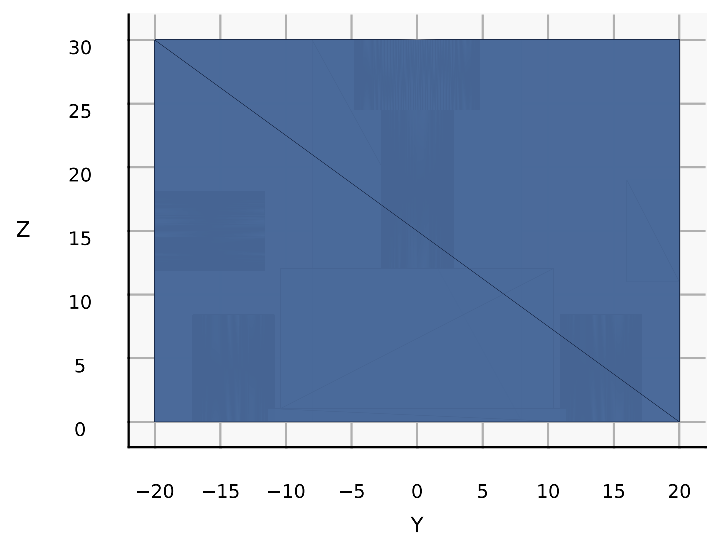
DIMENSIONS (mm)

Beam L x W x H	178.0 x 40.0 x 30.0
Mast spacing	150.0 (X=+/-75)
Mast pockets	20.8 sq x 12.0 deep
Mast bolt	M5, CB 9.5 x 5.5 on TOP
Pi mount PCD	58.0 x 28.0 (Pi 4 std)
Cable channel	8.0 x 4.0 (rear)
Insert pocket	6.2 dia x 8.5 deep
Front inserts	6 @ 22.0 pitch
Service open	50.0 x 16.0 top

Front View (-Y face) -- 6x accessory inserts



Side View (+X face) -- end profile + cable channel



ASSEMBLY

V1.1 changes from V1.0:
- Beam 30x25 -> 40x30 (fits Pi 4 PCD 49)
- Mast pockets 8 -> 12 mm (better torsion)
- Center service opening on TOP face
- Mast bolt CSK -> counterbore (socket-head)

ASSEMBLY:
1. Print PETG flat on bed, 100% infill, 4 walls, 0.2 mm layers
2. Heat 4x M5 inserts into BOTTOM Pi mount
3. Heat 6x M5 inserts into FRONT face
4. Drop beam over both mast tops (12 mm engage)
5. M5 x 30 socket-head through TOP counterbores self-tap into 2020 center bore
6. SD card / cables via center service opening
7. Mount Pi Carrier V1.0 to BOTTOM inserts
8. Route cables up REAR face cable channel

EXPANSION (6x M5, FRONT face):
- Status LED, mic, motion sensor, OLED, camera, future UI